LESSON PLAN

AIRCRAFT ARMAMENT (18)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Role of I-band transponder
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Role of I-band transponder. In ASW Operation.
5. Specific Objectives	18.1.4.1 Trainee will be able to appraise the function of secondary radars in ASW operation.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of I-band transponder function in ASW context using L & Ds and IB; Instructor supervision required for query resolution.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a transponder in aircraft systems?
(b) How does a secondary radar system differ from a primary radar system?
(c) Why is the I-band frequency preferred for transponder operations in ASW?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of I-band transponder • Interaction with ASW secondary radar • Data link protocols in ASW • Identification Friend or Foe (IFF) modes • Response mechanisms to interrogations	L & Ds	PPT, IB	15 min
(b)	• Role in target identification • Cooperation with ASW sonobuoys • Data fusion in ASW operations • Tracking and classification of contacts • Mitigation of false alarms	L & Ds	PPT, PDF	20 min
(c)	• Operational limitations in ASW • Interference management techniques • Maintenance implications for ASW readiness • Integration with combat management systems • Future trends in ASW radar tech	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the I-band transponder aids in distinguishing friendly from hostile aircraft during ASW operations.
(b) Describe the specific role of the transponder in data fusion with sonobuoy arrays.

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts learned in this lesson will be applied in the next lesson regarding the basic block diagram of the I-band transponder.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Explanation of basic block diagram of I-band transponder.	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min